# Sharma_DSC520Week11.2

2025-02-18

install.packages("class") install.packages("rattle")

## Set working directory for this week assignment data

```
## Pulling the Data frame for the binary-classifier data as below.
binary_classifier_data <- read.csv('binary-classifier-data.csv')

head(binary_classifier_data)
```

```
##   label        x        y
## 1     0 70.88469 83.17702
## 2     0 74.97176 87.92922
## 3     0 73.78333 92.20325
## 4     0 66.40747 81.10617
## 5     0 69.07399 84.53739
## 6     0 72.23616 86.38403
```

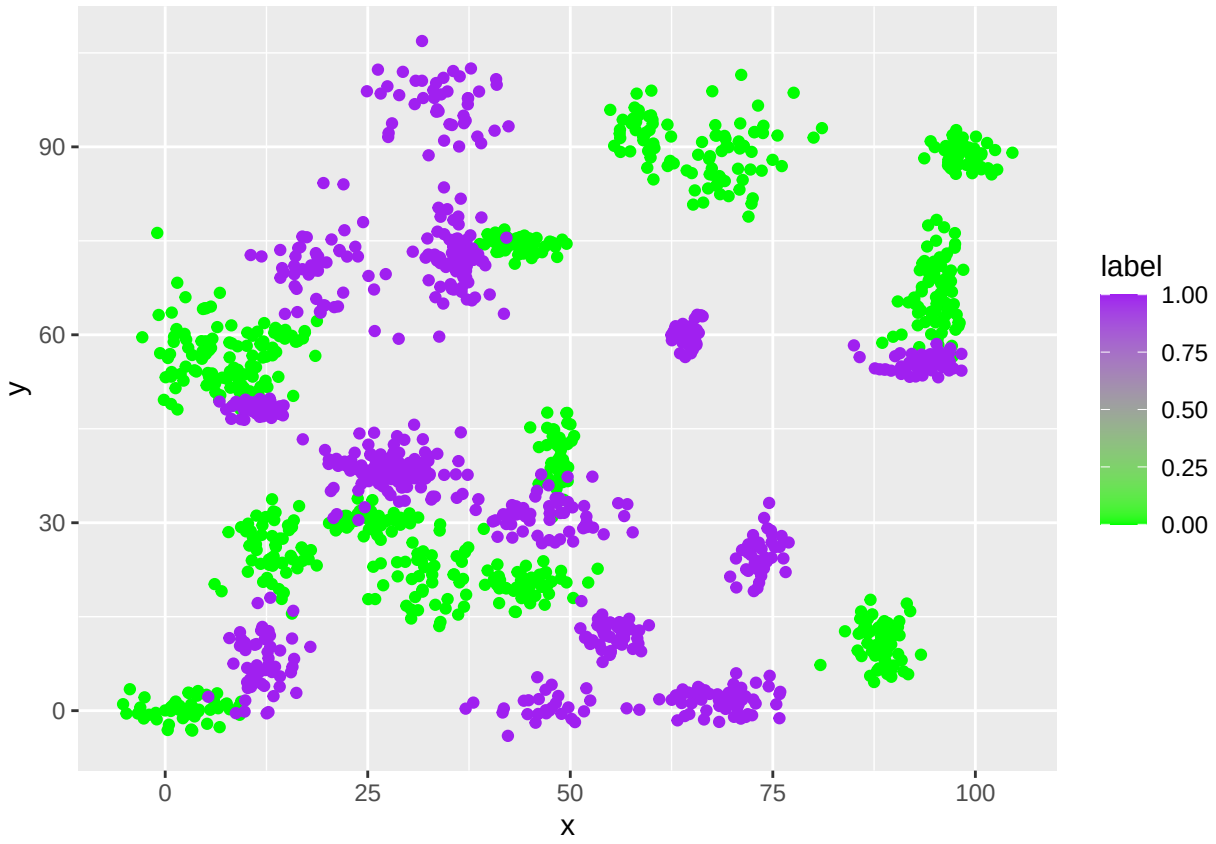
```
## Next I will be pulling the data for the trinary-classifier data as below.
trinary_classifier_data <- read.csv('trinary-classifier-data.csv')

head(trinary_classifier_data)
```

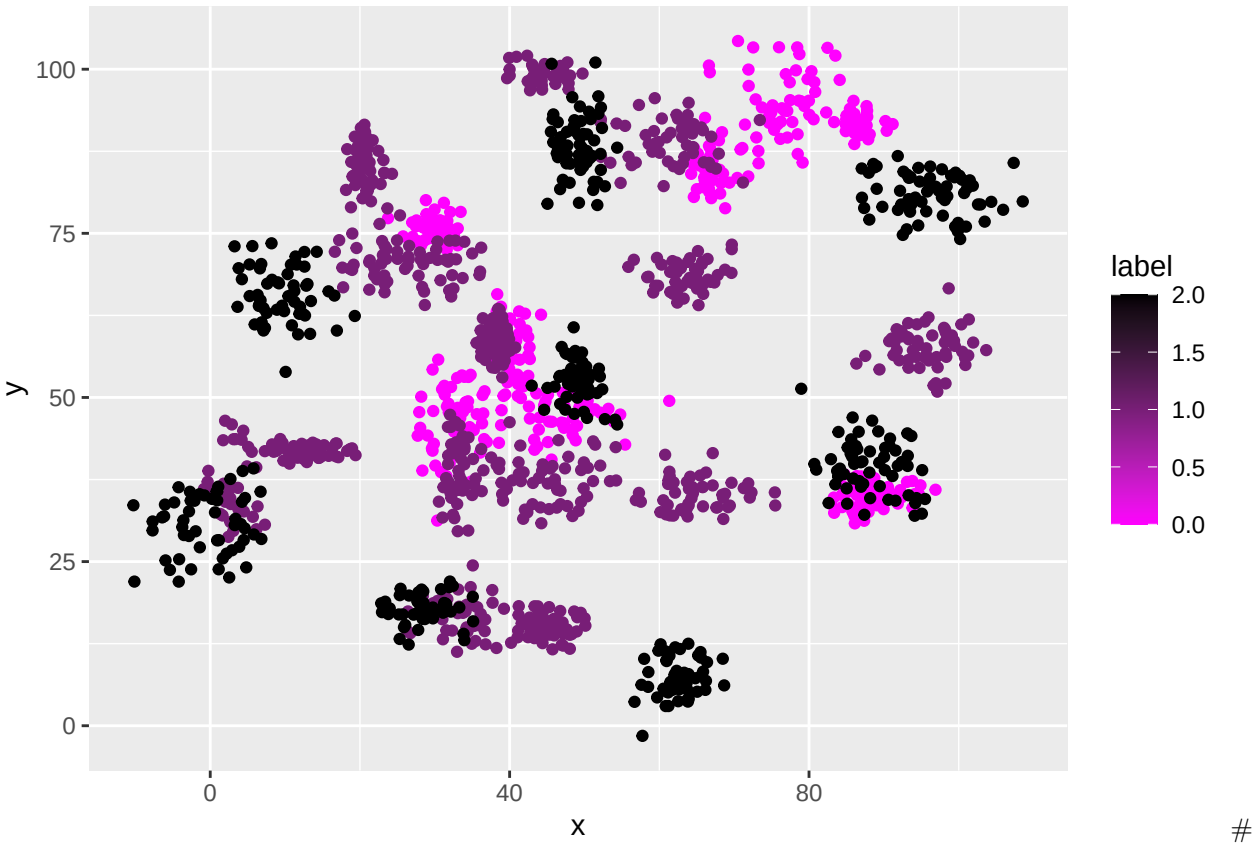
```
##   label        x        y
## 1     0 30.08387 39.63094
## 2     0 31.27613 51.77511
## 3     0 34.12138 49.27575
## 4     0 32.58222 41.23300
## 5     0 34.65069 45.47956
## 6     0 33.80513 44.24656
```

#Plot the data from each dataset using a scatter plot.

```
## Created a scatter plot for the binary_classifier_data frame
ggplot(binary_classifier_data,aes(x,y,color=label)) +
  geom_point() +
  scale_color_gradient2(low="white", mid="green", high="purple")
```




```
## Next I created a scatter plot for the trinary_classifier_data frame
ggplot(trinary_classifier_data, aes(x,y,color=label)) +
  geom_point() +
  scale_color_gradient2(low="white", mid="magenta", high="black")
```




# #

The k nearest neighbors algorithm categorizes an input value by looking at the # labels for the k nearest points and assigning a category based on the most # common label. In this problem, you will determine which points are nearest by # calculating the Euclidean distance between two points. As a refresher, # the Euclidean distance between two points:

```
# First I will create data normalization for both bindary and trinary data frames.
binary <- binary_classifier_data[, c("x", "y")]
trinary <- trinary_classifier_data[, c("x", "y")]

# Next I will create my train and test data sets for both binary and
# trinary data frames.train and test data sets for binary data and pull the
# number of observations for the binary train and test data
set.seed(150)
binary_selection<-sample(1:nrow(binary),size=nrow(binary)*0.60, replace = FALSE)
binary_train <- binary_classifier_data[binary_selection,]
NROW(binary_train)
```

```
## [1] 898
```

```
binary_test <- binary_classifier_data[-binary_selection,]
NROW(binary_test)
```

```
## [1] 600
```

```
# Next I will create dataframes for both train and test label data for the
# binary data and pull the number of observations for the binary train
train_label_binary <- binary_classifier_data[binary_selection,1,drop=TRUE]
NROW(train_label_binary)
```

```
## [1] 898
```

```
test_label_binary <- binary_classifier_data[-binary_selection,1,drop=TRUE]
NROW(test_label_binary)
```

```
## [1] 600
```

```
# train and test data sets for trinary data and pull the number of observations
# for the trinary train and test data
set.seed(130)
trinary_selection<-sample(1:nrow(trinary),
                          size=nrow(trinary)*0.60,replace = FALSE)
trinary_train <- trinary_classifier_data[trinary_selection,]
NROW(trinary_train)
```

```
## [1] 940
```

```
trinary_test <- trinary_classifier_data[-trinary_selection,]
NROW(trinary_test)
```

```
## [1] 628
```

```
# Next I will create dataframes for both train and test label data for the
# trinary data and pull the number of observations for the trinary and test data
train_label_trinary <- trinary_classifier_data[trinary_selection,1,drop=TRUE]
NROW(train_label_trinary)
```

```
## [1] 940
```

```
test_label_trinary <- trinary_classifier_data[-trinary_selection,1,drop=TRUE]
NROW(test_label_trinary)
```

```
## [1] 628
```

**Fit a k nearest neighbors' model for each dataset for k=3, k=5, k=10, k=15,**

**k=20, and k=25. Compute the accuracy of the resulting models for each value of**

**k.Plot the results in a graph where the x-axis is the different values of k and**

**the y-axis is the accuracy of the model.**

**Fitting a model is when you use the input data to create a predictive model.**

**There are various metrics you can use to determine how well your model fits**

**data. For this problem, you will focus on a single metric, accuracy.**

**Accuracy is simply the percentage of how often the model predicts the correct**

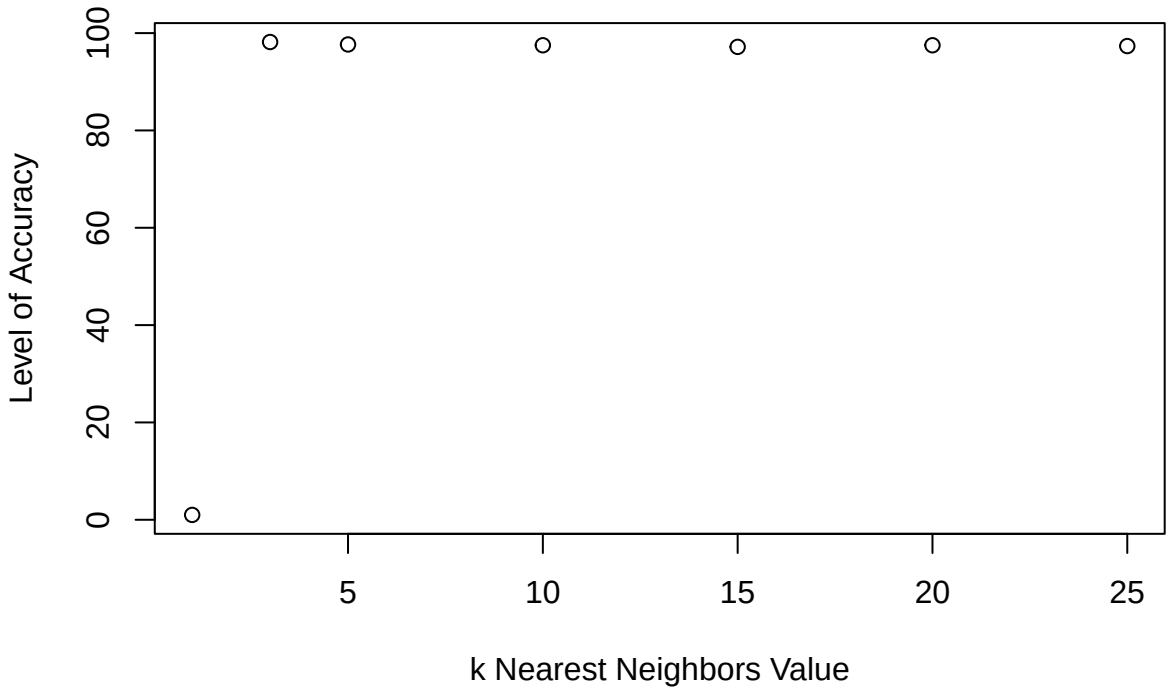
**result. If the model always predicts the correct result, it is 100% accurate.**

**If the model always predicts the incorrect result, it is 0% accurate.**

```
# Create a K nearest list the k nearest neighbors for binary data.
k_nearest <- list(3,5,10,15,20,25)
input=1
binary_accuracy=1
for (input in k_nearest)
  {
  nearest_neighbor <- knn(train = binary_train,
                          test = binary_test,
                          cl=train_label_binary,
                          k=input)
  binary_accuracy[input] <- 100*
    sum(test_label_binary == nearest_neighbor)/NROW(test_label_binary)
  k=input
  cat(k,'=',binary_accuracy[input],'')
}
```

```
## 3 = 98.16667 5 = 97.66667 10 = 97.5 15 = 97.16667 20 = 97.5 25 = 97.33333
```

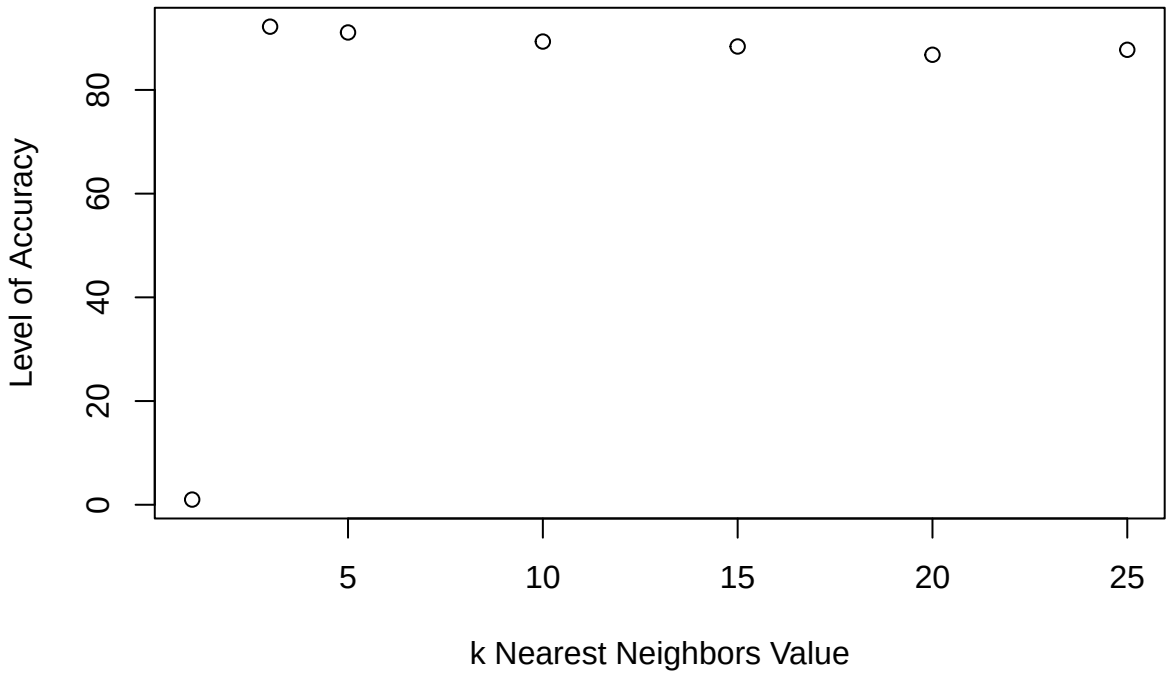
```
# Next I will plot the accuracy of my k values for binary
plot(binary_accuracy, type = "b", xlab = "k Nearest Neighbors Value",
ylab = "Level of Accuracy")
```




```
# Create a K nearest list the k nearest neighbors for trinary data.
k_nearest_trinary <- list(3,5,10,15,20,25)
input_trinary=1
trinary_accuracy=1
for (input_trinary in k_nearest_trinary)
  {
  trinary_nearest <- knn(train = trinary_train,
                         test = trinary_test,
                         cl=train_label_trinary,
                         k=input_trinary)
  trinary_accuracy[input_trinary] <- 100*
    sum(test_label_trinary == trinary_nearest)/NROW(test_label_trinary)
  k=input_trinary
  cat(k,'=',trinary_accuracy[input_trinary])
}
```

```
## 3 = 92.197455 = 91.082810 = 89.3312115 = 88.375820 = 86.7834425 = 87.73885
```

```
# Next I will plot the accuracy of my k values for binary
plot(trinary_accuracy, type = "b",
      xlab = "k Nearest Neighbors Value",
      ylab = "Level of Accuracy")
```




```
  k=input_trinary
  cat(k,'=',trinary_accuracy[input_trinary])
```

```
## 25 = 87.73885
```

**Looking back at the plots of the data, do you think a linear classifier would**

**work well on these datasets? yes I believe so as a linear classifier can**

**classify data into labels based on a linear combination of input features**

**How does the accuracy of your logistic regression classifier from last week**

**compare? Why is the accuracy different between these two methods?**

**The accuracy from this weeks method shows that for our trinary data that our**

**accuracy went down as the k value went up as seen in our graph and outputs.**

**While in our binary data we saw that as the k value increased the accuracy**

**decreased as well but at a higher scale.**

```
## Pulling the Data frame for the clustering_data as seen below.
clustering_data <- read.csv('clustering-data.csv')
head(clustering_data)
```

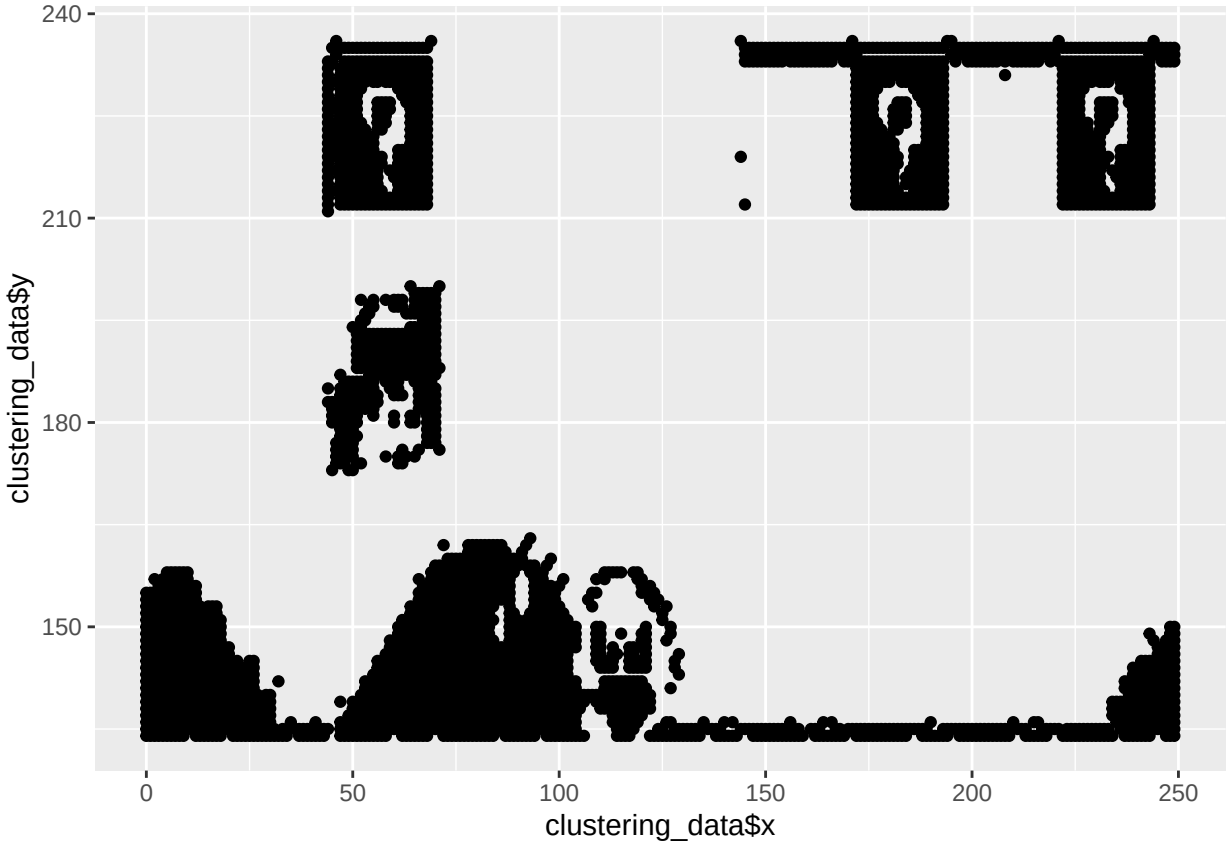
```
##     x   y
## 1  46 236
## 2  69 236
## 3 144 236
## 4 171 236
## 5 194 236
## 6 195 236
```

**Plot the dataset using a scatter plot.**

```
## Created a scatter plot for the binary_classifier_data frame
ggplot(clustering_data, aes(x=clustering_data$x, y=clustering_data$y)) +
  geom_point()
```

```
## Warning: Use of `clustering_data$x` is discouraged.
## i Use `x` instead.
```

```
## Warning: Use of `clustering_data$y` is discouraged.
## i Use `y` instead.
```




# Fit the dataset using the k-means algorithm from k=2 to k=12. # Create a scatter plot of the resultant clusters for each value of k.

```
head(clustering_data)
```

```
##     x   y
## 1  46 236
## 2  69 236
## 3 144 236
## 4 171 236
## 5 194 236
## 6 195 236
```

```
# standardize the variable
clust_standard <- scale(clustering_data[-1])
# k-means from k=2 to k=12
clust_k_means_fit <- kmeans(clust_standard, 4)
# collect attributes
attributes(clust_k_means_fit)
```

```
## $names
## [1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss"
## [6] "betweenss"    "size"         "iter"         "ifault"
##
## $class
## [1] "kmeans"
```

```
# view the centroids
clust_k_means_fit$centers
```

```
##           y
## 1 -0.6363489
## 2  1.2575905
## 3 -0.9703621
## 4  0.2920086
```

```
# view the clusters
clust_k_means_fit$cluster
```

```
##    [1] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
##   [38] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
##   [75] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
##  [112] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
##  [149] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
##  [186] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
##  [223] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
##  [260] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
##  [297] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
##  [334] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
##  [371] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
##  [408] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
##  [445] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
##  [482] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
##  [519] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
##  [556] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
##  [593] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
##  [630] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
##  [667] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
##  [704] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
##  [741] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
##  [778] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
##  [815] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
##  [852] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
##  [889] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
##  [926] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
##  [963] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
## [1000] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
## [1037] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
## [1074] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
## [1111] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
## [1148] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
## [1185] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
## [1222] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
## [1259] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
## [1296] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
## [1333] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
## [1370] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
## [1407] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
## [1444] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
## [1481] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
## [1518] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
## [1555] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
## [1592] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
## [1629] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
## [1666] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
```

```
## [1703] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4
## [1740] 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 1 1 1 1 1 1 1 1 1
## [1777] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [1814] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [1851] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [1888] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [1925] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [1962] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [1999] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2036] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2073] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2110] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2147] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2184] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2221] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2258] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2295] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2332] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2369] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2406] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2443] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2480] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2517] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2554] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2591] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [2628] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [2665] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [2702] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [2739] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [2776] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [2813] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [2850] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [2887] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [2924] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [2961] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [2998] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [3035] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [3072] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [3109] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [3146] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [3183] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [3220] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [3257] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [3294] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [3331] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [3368] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [3405] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [3442] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [3479] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [3516] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [3553] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [3590] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [3627] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [3664] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
```

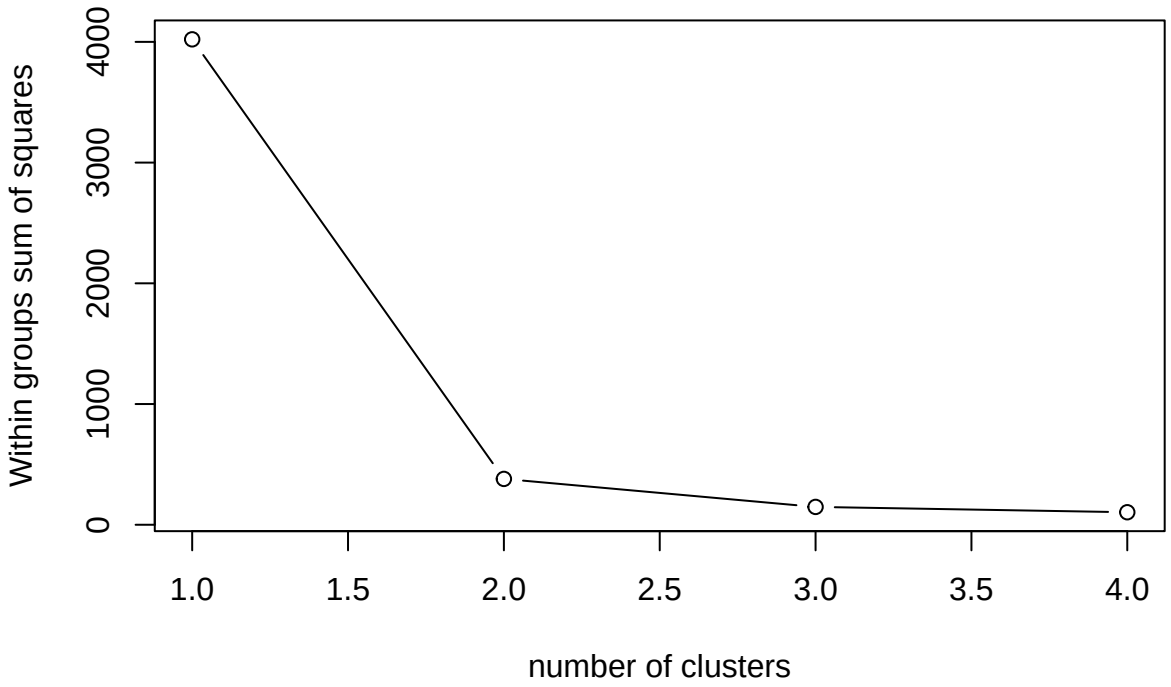
```
## [3701] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [3738] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [3775] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [3812] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [3849] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [3886] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [3923] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [3960] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## [3997] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
```

```r
# create the cluster
clust_nearest <- list(2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12)
clust_input = 1
seed = 1234
nc = 12
clustplot <- function(clustering_data, nc=12, seed=1234)
  {
  clustss <- (nrow(clustering_data)-1)*sum(apply(clustering_data,2,var))

for (clust_input in 2:nc)
  {
   set.seed(seed)
   clustss[clust_input] <-sum(kmeans(clustering_data,
                                     centers = clust_input)$withinss)
}

# Plot my cluster using the plot function
plot(1:nc, clustss, type="b",
     xlab="number of clusters",
     ylab="Within groups sum of squares")
}
```

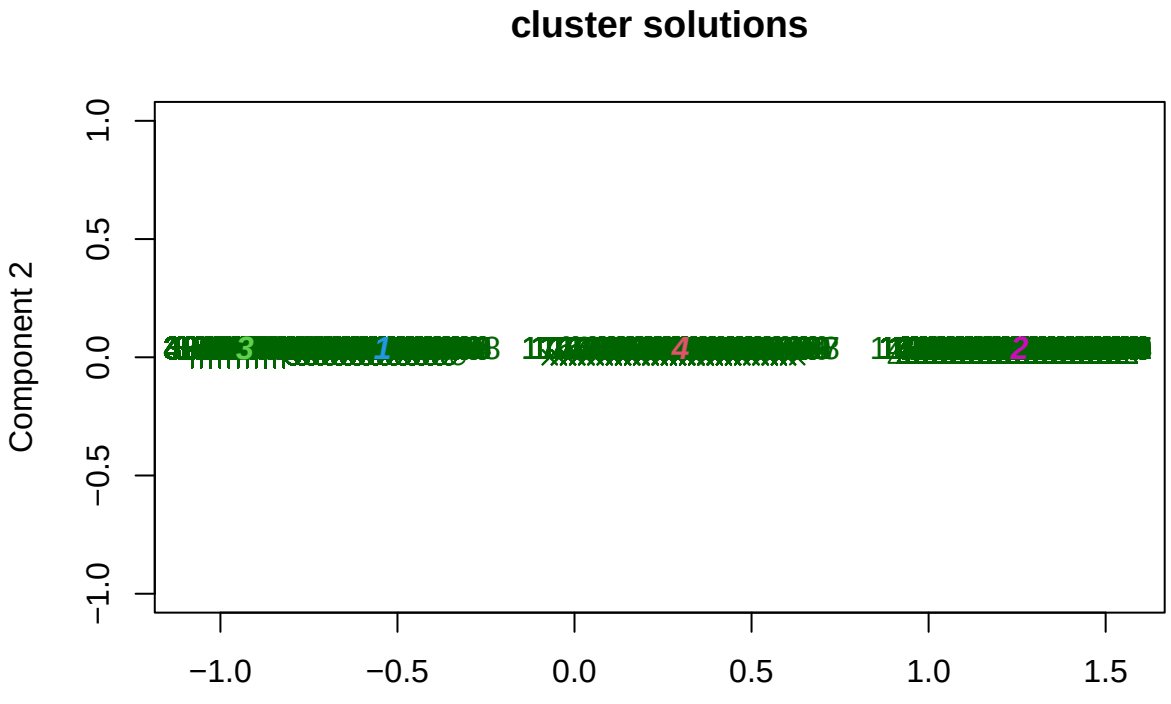
```r
# As seen in our graph that was generated in the previous example we see that
# our elbow point was at 4
clustplot(clust_standard, nc =4)
```




# Next

I will create a cluster plot to see the best value

```
# use the function clustplot to plot our findings
clusplot(clust_standard,
         clust_k_means_fit$cluster,
         main = 'cluster solutions',
         color = TRUE, shade = TRUE, labels = 2, lines = 0)
```




# As k-

means is an unsupervised algorithm, you cannot compute the accuracy as # there are no correct values to compare the output to. Instead, you will use the # average distance from the center of each cluster as a measure of how well the # model fits the data. To calculate this metric, simply compute the distance of # each data point to the center of the cluster it is assigned to and take the # average value of all of those distances.

```
# I will calculate the distance between and plot
clust_dist <- dist(clust_standard)
# create hclust for different methods
clust_complete <- hclust(clust_dist, method = "complete")
clust_single <- hclust(clust_dist, method = "single")
clust_average <- hclust(clust_dist, method = "average")
# plot data
par(mfrow= c(1,3))
plot(clust_complete, main = 'Complete link')
plot(clust_single, main = 'Single link')
plot(clust_average, main = 'Average link')
```